Machine Learning Case Studies

KNN, Linear Regression and El Niño / La Niña

Mushroom classification · Bike-sharing demand · ENSO forecasting

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Case study 01

K-Nearest Neighbours

Classifying mushrooms as edible or poisonous from 22 categorical features.

What is KNN?

A **supervised** algorithm for both classification and regression.

It works on **proximity**: a new point is predicted from the labels or values of its nearest neighbours in the training set.

Called a **lazy learner** — it stores the whole dataset and only computes at prediction time.

	Classification	Regression
Output	Class / category	Number
Example	Cat / Dog	House price
KNN uses	Majority voting	Averaging
Neighbours	A, A, B, A, B	10, 20, 30
Prediction	A	20

How KNN works

1 · Choose K

2 · Calculate distance
Euclidean (most common), Manhattan, Minkowski

3 · Sort & pick the K nearest

4 · Vote or average
Majority vote for classification; mean for regression

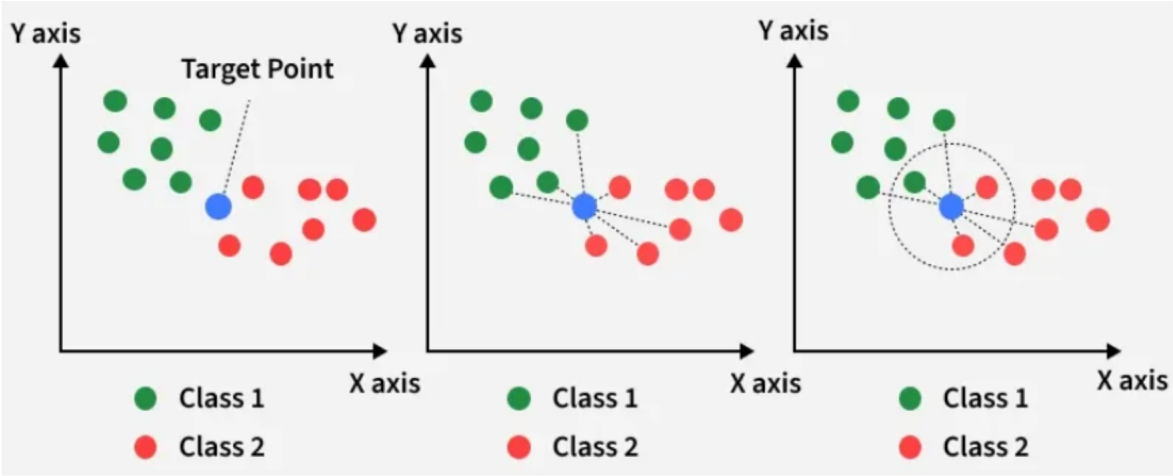


Figure: the target point takes the class of its K closest neighbours.

Choosing the right K

Larger K smooths out noise and outliers — but too large a K makes the model too simple and it **underfits**.

Cross-validation

Split data into parts; train on some, test on the rest, and compare K values.

Elbow method

Plot accuracy against K; error drops at first, then stops improving.

Odd values of K

Use an odd K in classification to avoid tied votes.

Dataset: Mushroom Classification

8,124
samples

22
categorical features

51.8%
edible (4,208)

48.2%
poisonous (3,916)

Source

23 species of gilled mushrooms (Agaricus & Lepiota), Audubon Society Field Guide, 1981.

Preprocessing

- No duplicates; “?” in stalk-root → 2,480 missing values
- Most-frequent imputation + One-Hot Encoding in one pipeline
- 80:20 stratified split → 6,499 train / 1,625 test

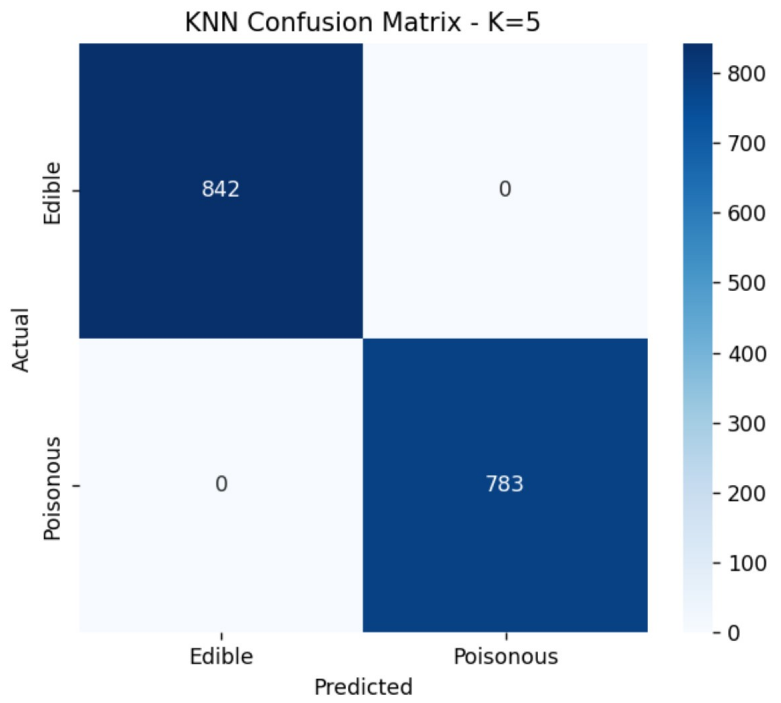
Baseline model: K = 5, Euclidean

100%

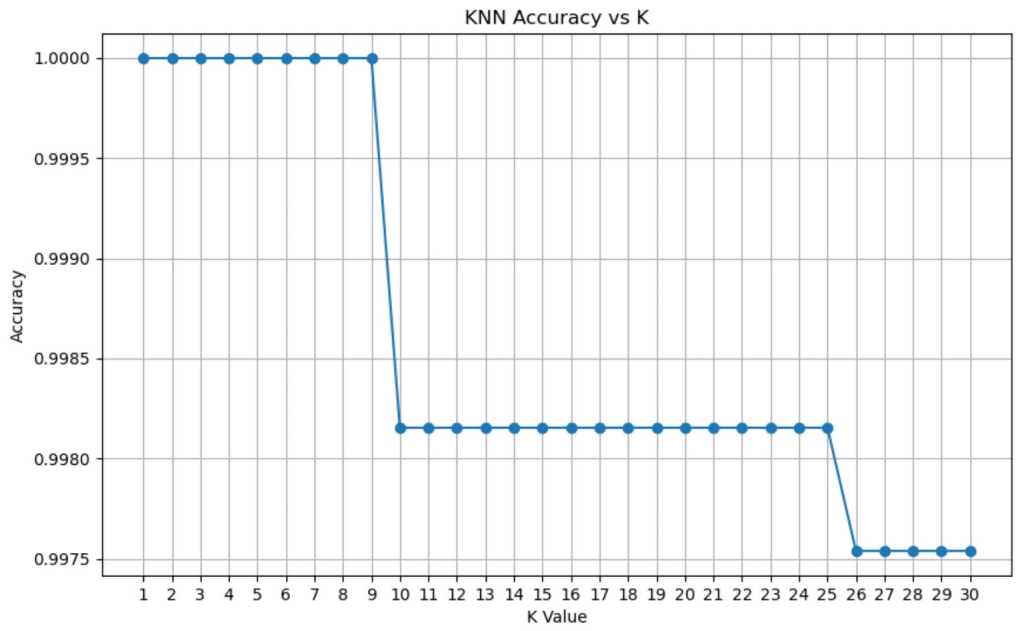
test accuracy on 1,625 unseen samples

Class	Precision	Recall	F1	n
Edible	1.00	1.00	1.00	842
Poisonous	1.00	1.00	1.00	783

Preprocessing sits inside the pipeline, so it is learned only from training data.



Accuracy across K = 1 … 30



K	Test accuracy
1 – 9	100%
10 – 25	99.82%
26 – 30	99.75%

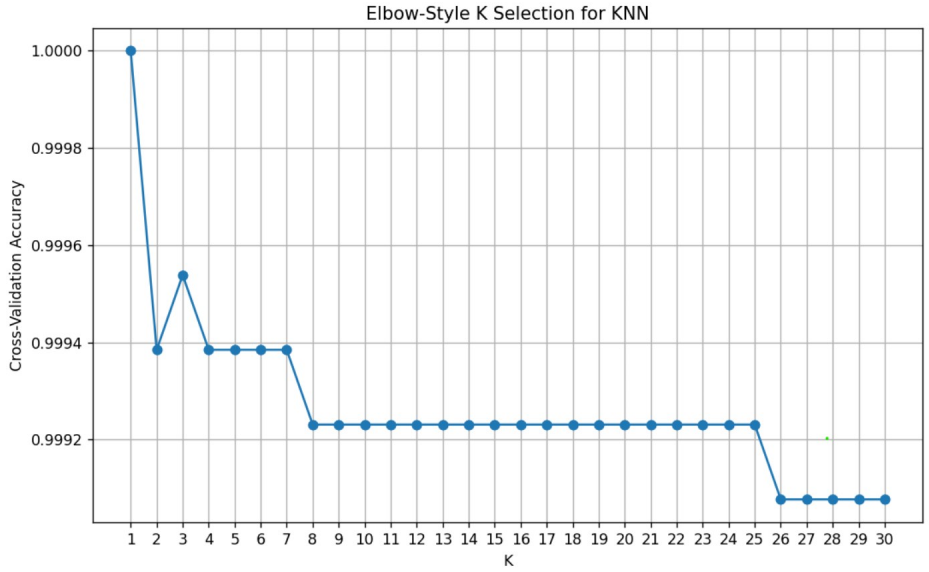
Several K values tie at 100%, so K = 1 is **not uniquely optimal** — the code simply picks the first one.

Comparing K-selection methods

Method	Selected K	Test accuracy
Cross-validation	1	100%
Elbow method	2	100%
Odd K	1	100%

- 5-fold CV: K = 1 scores 1.0000 (std 0.0); K = 3 next at 0.9995
- Odd K 1–29: all above 99.9%
- Elbow at K = 2, after the sharp drop from K = 1

All three choices give perfect test accuracy on this dataset.



Case study 02

Linear Regression

Predicting hourly bike-sharing demand in Washington, D.C.

What is Linear Regression?

A **supervised** algorithm that predicts a **continuous** output from one or more input features.

It fits the straight line (or hyperplane) that best matches the data.

Fast, interpretable, and the usual **baseline** before more complex models.

	Simple	Multiple
Inputs	One variable	Two or more
Model	$\hat{y} = \theta_0 + \theta_1 x$	$\hat{y} = \theta_0 + \theta_1 x_1 + \ldots + \theta_n x_n$
Geometry	Straight line	Hyperplane
Example	Rentals from temp	Rentals from temp, humidity, hour, …

How Linear Regression works

1 · Hypothesis

$\hat{y} = \theta_0 + \theta_1 x$

2 · Residual

$e_i = y_i - \hat{y}_i$

3 · Cost (half MSE)

$J = (1/2m) \Sigma (\hat{y}_i - y_i)^2$

4a · Ordinary Least Squares

$\theta_1 = \Sigma(x - \bar{x})(y - \bar{y}) / \Sigma(x - \bar{x})^2$
$\theta_0 = \bar{y} - \theta_1 \bar{x}$ — exact, one step

4b · Gradient Descent

$\theta_j := \theta_j - \alpha \cdot \partial J / \partial \theta_j$
Iterative; scales to large datasets

Step 5: Evaluating the model

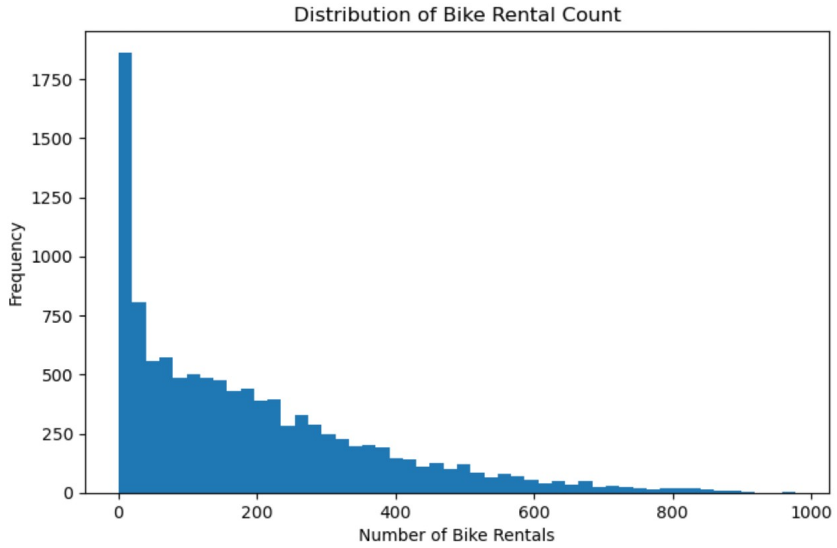
Metric	Formula	Meaning
MAE	$(1/n)\ \Sigma\lvert y - \hat{y}\rvert$	Average absolute error, in target units
MSE	$(1/n)\ \Sigma(y - \hat{y})^2$	Average squared error; penalises large errors
RMSE	$\sqrt{\text{MSE}}$	Error in target units
R^2	$1 - SS_{res} / SS_{tot}$	Share of variance in y explained
Adjusted R^2	$1 - (1 - R^2)(n - 1)/(n - p - 1)$	R^2 corrected for p features

Dataset: Bike Sharing Demand

10,886
hourly records, 2011–12

191.6
mean rentals / hour (median 145)

- No missing values or duplicates
- Extracted year, month, day, hour, weekday from datetime
- Dropped **casual** & **registered** — they sum to count (data leakage)



Target is right-skewed: most hours are quiet, a few reach 977.

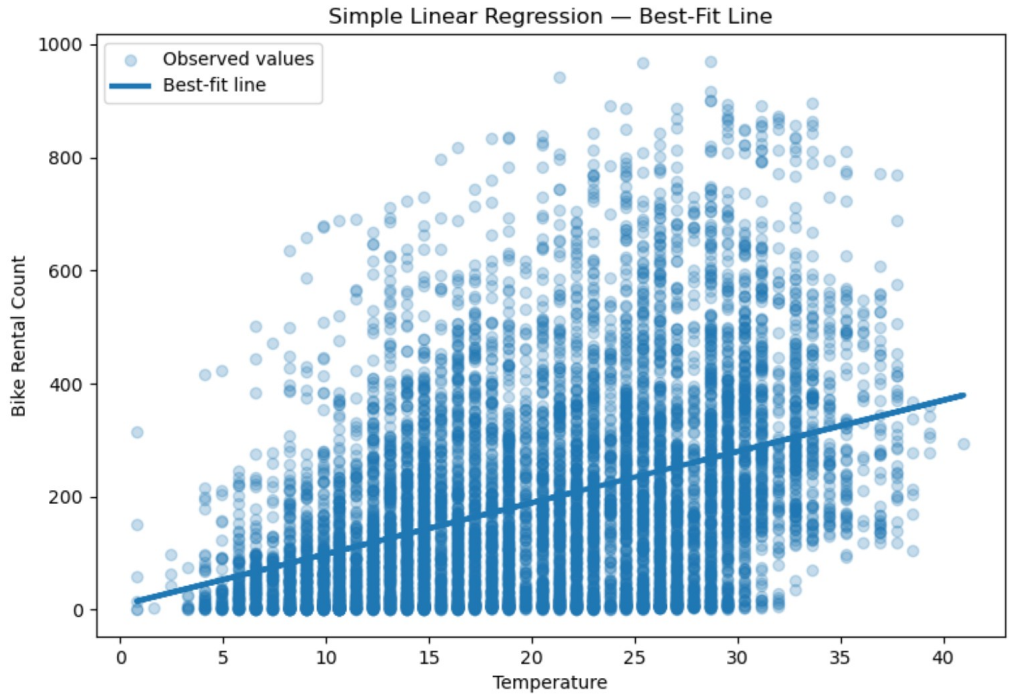
Simple model: rentals from temperature

$$\hat{y} = 7.8550 + 9.0645 \times \text{temp}$$

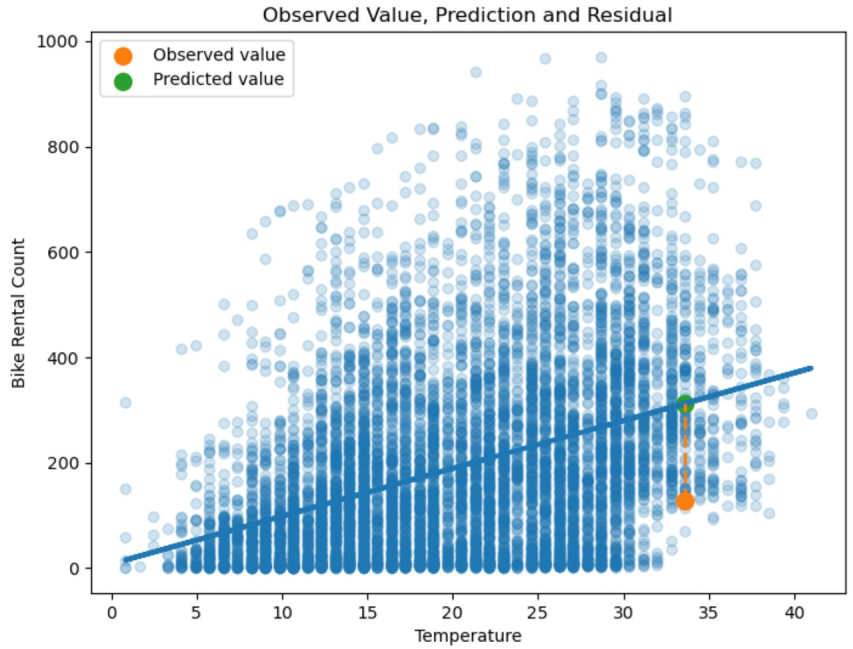
Each 1 °C adds about **9 rentals per hour**.

80:20 split → 8,708 train / 2,178 test.

The line captures the upward trend, but points scatter widely — temperature alone leaves much unexplained.



Observed, predicted and residual



First test sample	Value
Observed y	127
Predicted ŷ	312.60
Residual y − ŷ	−185.60

Negative residual → the model **over-predicted** this hour by ~186 rentals.

Worst of the first 10: actual 601, predicted 104 (error 496.5).

Three routes, one answer

9.064522
slope from scikit-learn and manual OLS — identical

189.15
h(20 °C): predicted rentals, close to the 191.57 mean

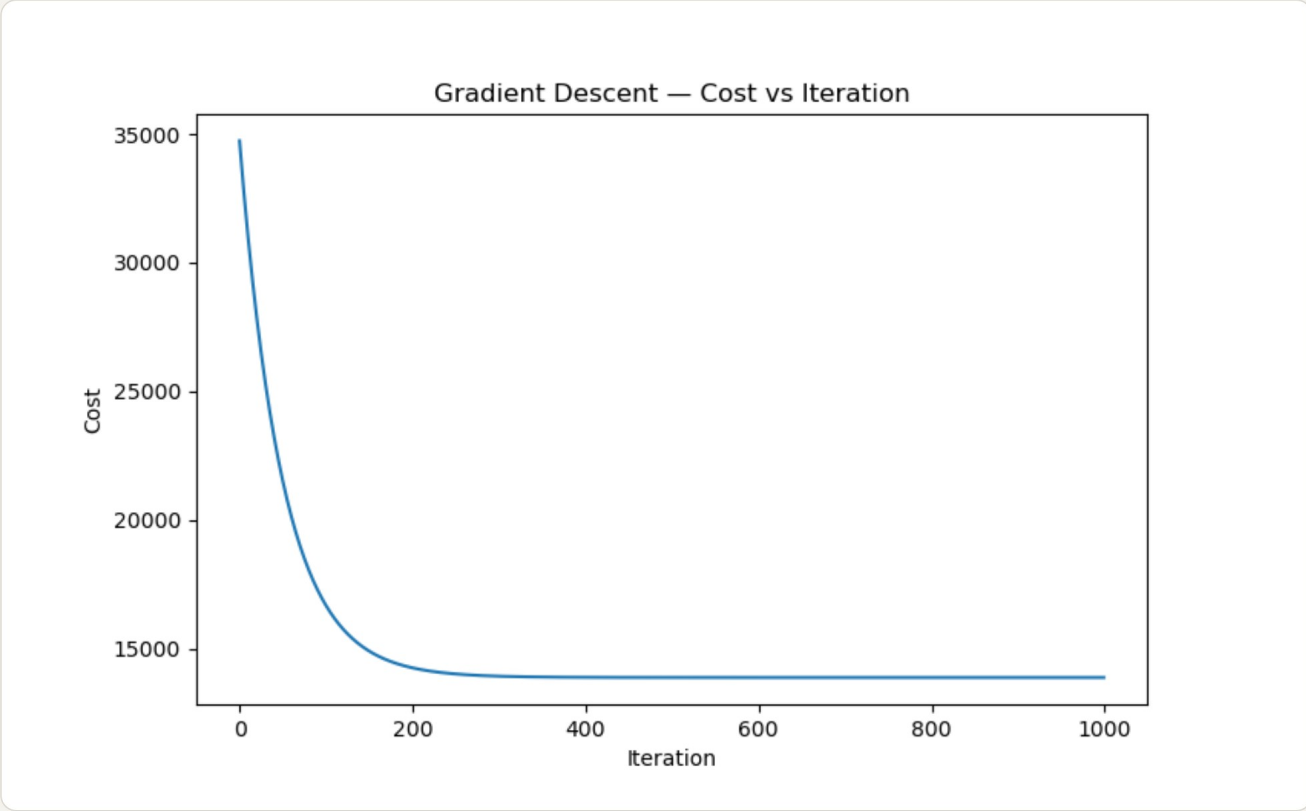
13,888.78
minimum cost J (training RMSE ≈ 166.7)

Manual least squares matched scikit-learn to six decimal places, confirming LinearRegression uses the OLS solution. An RMSE of ~167 against a mean of ~192 shows temperature is a weak predictor on its own.

Gradient descent converges to the same line

Setting	Value
Learning rate α	0.01
Iterations	1,000
Start θ_0, θ_1	0, 0
Final θ_0	191.58
Final θ_1	70.59

Temperature was standardised first. Cost falls from ~34,700 to 13,889 — the same minimum as OLS.



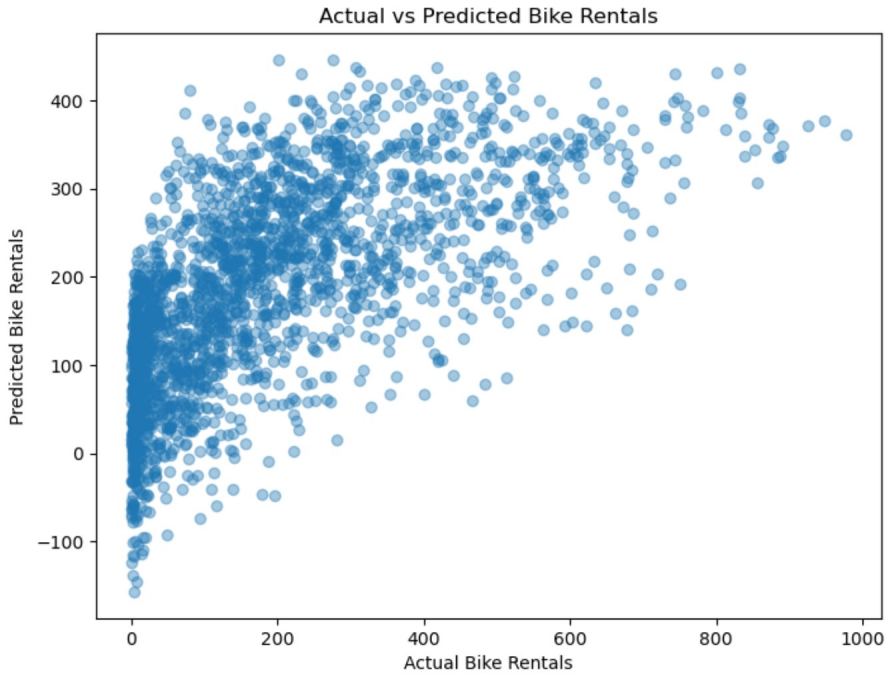
θ_1 = 70.59 is per standard deviation; 70.59 / 9.0645 ≈ 7.79 °C, the std of temperature.

Multiple regression: 17 features

0.40
R^2 on test set

140.6
RMSE (MAE 105.1)

- 13 inputs; season and weather one-hot encoded (drop_first) → 17 features
- Never predicts above ~450, missing peak demand of 600–977
- Some negative predictions (down to ~−150)
- Hour treated as linear, but demand peaks at commute times



Simple vs multiple — and what next

Aspect	Simple	Multiple
Features	1 (temp)	17
Key result	Train RMSE ≈ 166.7	Test RMSE 140.63, R^2 0.40
Strength	Easy to interpret	Uses weather, calendar and time
Limitation	Ignores other factors	Misses non-linear hourly peaks

Next steps: one-hot encode hour, log-transform the skewed target, add interaction/polynomial terms, or try tree-based models.

Case study 03

El Niño & La Niña

Forecasting the warm and cold phases of ENSO with machine learning.

What are El Niño and La Niña?

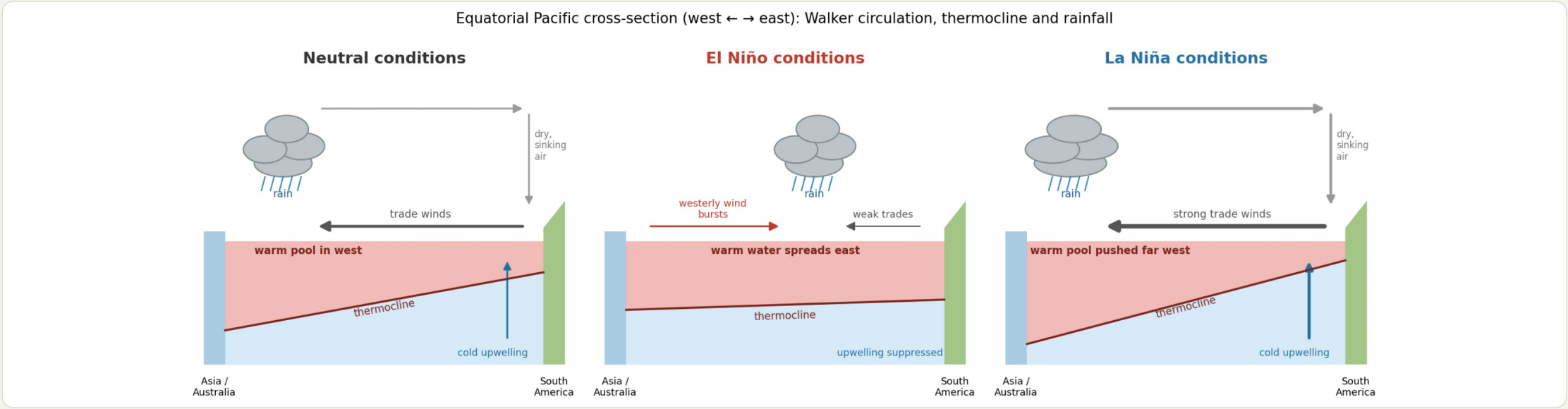
Warm and cold phases of the **El Niño–Southern Oscillation (ENSO)** in the tropical Pacific — a coupled ocean–atmosphere cycle.

Named by Peruvian fishermen who noticed warm coastal water around Christmas.

For India, El Niño years are closely watched for their link to weaker monsoons.

	El Niño	La Niña
Pacific SST	≥ +0.5 °C	≤ −0.5 °C
Trade winds	Weaken / reverse	Strengthen
Indian monsoon	Often weaker	Normal or above
Australia	Drier, bushfires	Wetter, floods
Duration	9–12 mo, every 2–7 yr	1–3 years

How ENSO works



Neutral

Trade winds push warm water west; cold upwelling in the east (Walker circulation).

El Niño

Winds weaken; warm water flows east, thermocline flattens, rain shifts east.

La Niña

Winds strengthen; more warm water piles west, east gets colder and drier.

Feedback and measurement

Bjerknes feedback

Warmer east → weaker trade winds → even warmer east.
Slow Kelvin and Rossby waves eventually reverse the anomaly.

Current situation

NOAA (10 Sep 2026): El Niño Advisory in effect; Niño 3.4 ≈ +1.8 °C, >90% chance of a very strong event in winter 2026–27.

Index	Used for
Niño 3.4	Main official ENSO index
ONI	NOAA's El Niño / La Niña definition
Niño 1+2	Coastal El Niño — used in this study
SOI	Atmospheric side (Tahiti − Darwin)
RONI	Separates ENSO from global warming

Machine learning for ENSO

Model	How it is used	Strength / weakness
Linear / ridge	Niño index from lagged values	Interpretable baseline; linear only
Random forest, XGBoost	Lagged indices, wind, heat content	Non-linear; needs hand-made features
CNN	Spatial patterns in SST & heat-content maps	Ham et al. 2019: skill ~1.5 years ahead
LSTM / ConvLSTM	How patterns evolve in time	Temporal memory; data hungry
Transformer	Attention over 3D ocean fields	Zhou & Zhang 2023: long-lead skill
Graph NN	Ocean grid as connected nodes	Captures teleconnections

Research highlight and challenges

CNN multi-year forecasts (Ham, Kim & Luo, 2019)

- Trained on CMIP5 simulations, fine-tuned on observations (transfer learning)
- Correlation skill > 0.5 up to ~17 months ahead
- Heat maps reveal ocean precursor regions

Challenges

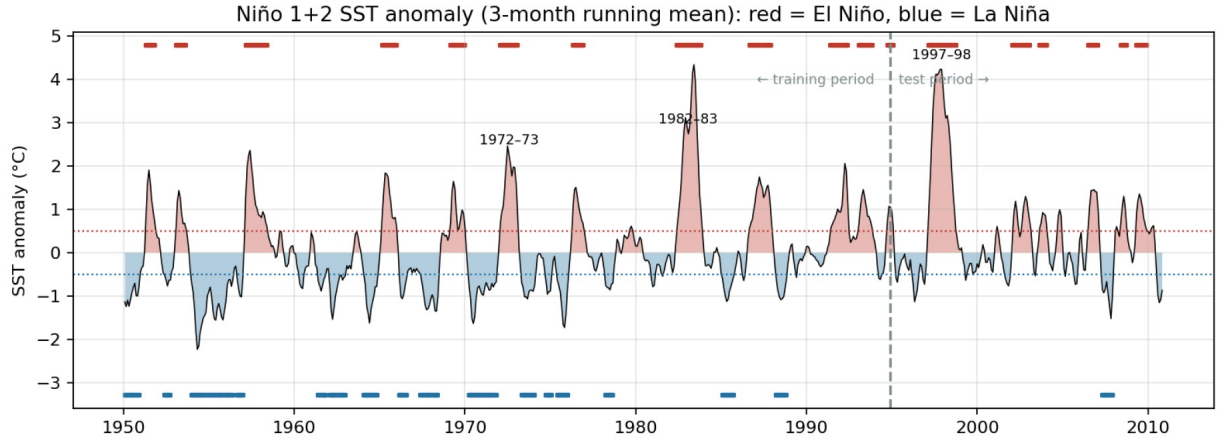
- Small data — only a few dozen events
- Spring predictability barrier (Mar–May)
- Non-stationarity from climate change
- Data leakage from careless preprocessing
- Need for interpretability

Dataset: Niño 1+2 sea surface temperature

- NOAA ERSST v3b via statsmodels, 1950–2010
- 732 monthly values, 0 missing; mean 23.09 °C
- Climatology from training years only (1950–94) → anomalies
- ONI-style labels: ±0.5 °C for ≥ 5 months

19
El Niño events

17
La Niña events



Top events	Peak
El Niño 1982–83	+4.34 °C
El Niño 1997–98	+4.24 °C
La Niña 1954–56	−2.24 °C

Features and models

15 features per month

- Anomaly lags 0–11
- sin / cos of month (seasonality)
- 3-month trend

Targets 1–12 months ahead. Chronological split: train 1950–94, test 1995–2010 — including the unseen 1997–98 super El Niño.

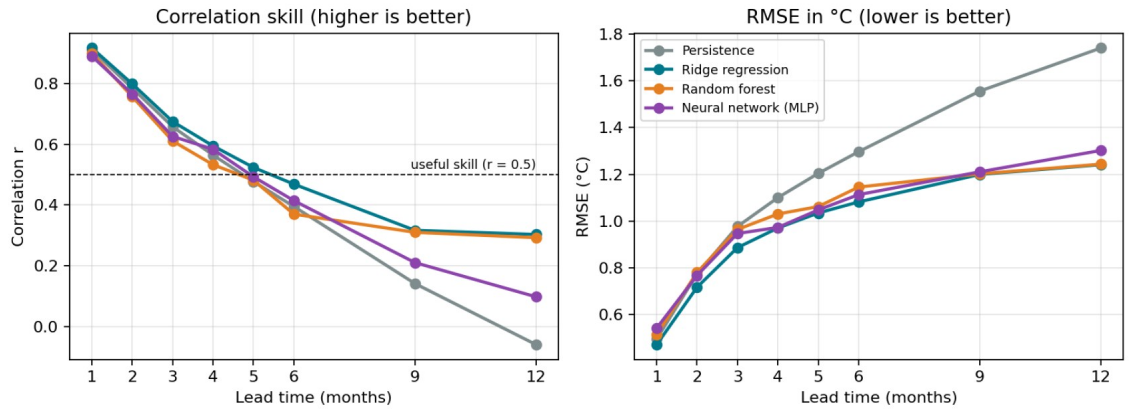
Four models

- **Persistence** — forecast = current anomaly
- **Ridge** — StandardScaler + α = 1.0
- **Random forest** — 400 trees
- **MLP** — layers (32, 16), early stopping

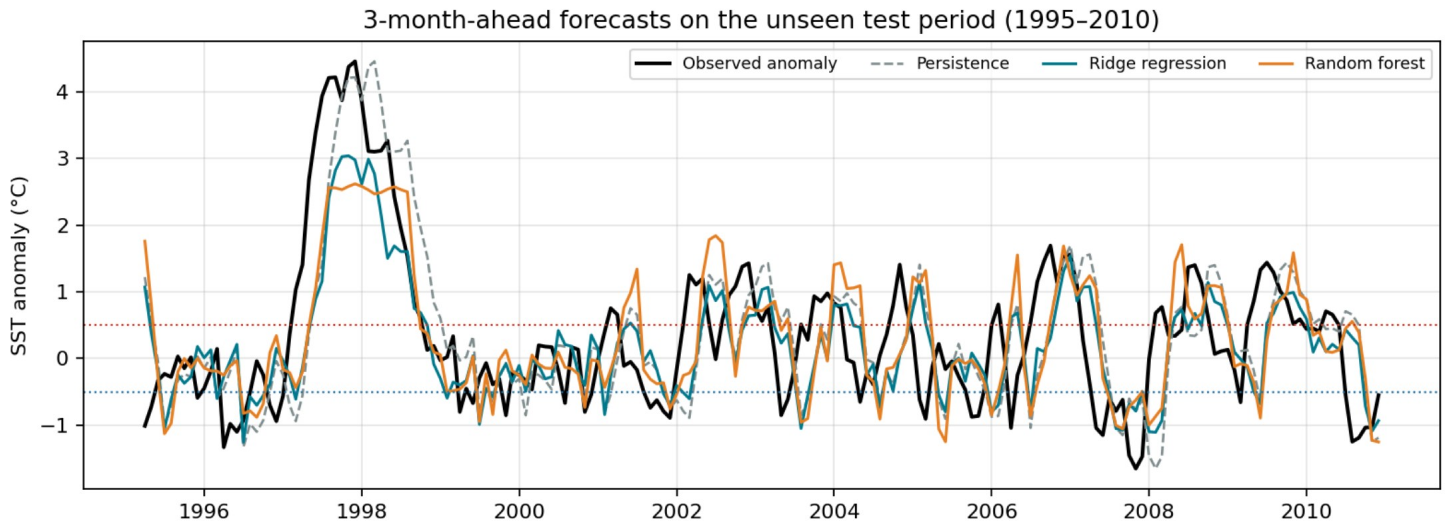
Forecast skill vs lead time

Lead	Persistence	Ridge	RF	MLP
1 mo	0.91	0.92	0.90	0.89
3 mo	0.66	0.68	0.61	0.63
6 mo	0.40	0.47	0.37	0.41
12 mo	−0.06	0.30	0.29	0.10

Correlation r shown. **Ridge** is best overall: at 12 months it cuts RMSE from 1.74 to 1.24 °C.



3-month forecasts on the unseen test period



Models track the 1997–98 El Niño, 2007–08 La Niña and 2009–10 El Niño.

But all **underestimate** the 1997–98 peak: +4.46 °C observed vs +3.04 °C (ridge).

Top RF feature: current anomaly (0.63), then lags 5–7 and trend.

Phase classification: El Niño / Neutral / La Niña

Method	Acc. 3 mo	F1 3 mo	Acc. 6 mo	F1 6 mo
Persistence	58.5%	0.48	44.9%	0.33
Ridge + threshold	65.4%	0.54	62.2%	0.47
Random forest	64.4%	0.54	56.8%	0.45

Both ML approaches beat persistence. At 3 months, ridge caught 38 of 59 El Niño months; most errors were Neutral months misread as La Niña (24) or El Niño (17). Only 8 La Niña months in the test set make those results noisy.

Findings

ML beats persistence

6-month RMSE 1.30 → 1.08 °C; r = 0.30 kept at 12 months.

Simple can win

With ~45 years of one index, ridge matched or beat RF and MLP.

Extremes are hard

Every model underestimated the 1997–98 super El Niño.

Preprocessing matters

No future data in anomalies or labels; split chronologically.

Physics + ML

Key features match known ENSO memory; heat content and winds are next.

Future scope

Niño 3.4, newer events (2015–16, 2023–24, 2026), CNN / ConvLSTM, CMIP6 transfer learning.

Across the three studies

KNN

100% test accuracy on mushrooms for every K-selection method.

Linear Regression

R^2 0.40 on bike demand — a solid baseline that misses non-linear peaks.

El Niño / La Niña

Ridge regression beats persistence at long leads; extremes stay hard.

Thank you — questions?